Chapter 1

THEORY OF METAL

Elementary Quantum Mechanical Ideas

Quantum mechanics is a generalized form of mechanics applicable to very small objects like electron, neutrons etc. in the atom. Classical mechanics are mainly applicable to large objects.

Various phenomena and process that take place within a material cannot be explained without having a good knowledge on quantum mechanics. Although classical mechanics gives ideas about various behavior of different objects, it fails to explain the material behavior for microscopic objects.

Q. Do sub-atomic particle obey Newton's laws of motion?
A. The behavior of the sub-atomic particle cannot be described by Newton's laws.

Wave Particle Duality: De Broglie's Equation

When we deal about sub-atomic particles like electron, photon, neutron we experience that these sub-atomic particle do not follow Newton's laws of motion as well as they do not obey the Maxwell's Electromagnetic Theory as wave. This is because these sub-atomic particles behave as both wave as well as particle.

It was observed that electromagnetic waves eg. X-rays, gamma rays, visible light etc. exhibit as similar to properties of discrete particles of matter.
In 1924, Louis de Broglie presented his research thesis, in which he proposed electrons have properties of both waves and particles, like light. The de Broglie equation is an equation used to describe the wave properties of matter, specifically, the wave nature of the electron:

$$\boldsymbol{\lambda = h/p}$$
$$\boldsymbol{\lambda = \frac{h}{mv}}$$

where λ is wavelength, p is momentum of a particle, h is Planck's constant, m is the mass of a particle, moving at a velocity v.

Also, $\boldsymbol{p = \hbar k}$ and $\hbar = \boldsymbol{h/2\pi}$

From above equations

$\boldsymbol{k = 2\pi/\lambda}$ is a wave number.

The value of h = 6.624×10^{-34} Js

The electromagnetic radiation energy possessed by the photons in the form of packets is given by,

$\boldsymbol{E = hf}$ (known as Plank's equation)

Also we know,

$\hbar = \boldsymbol{h/2\pi}$

Thus, above equation can be written as,

$$E = \hbar \times 2\pi f$$
$$E = \hbar\omega \text{ (ω is known as angular frequency)}$$

There were two experiments conducted in early 20th century which were un explained by classical mechanics and so warranted the need for quantum mechanical ideas.

Experiment 1

It was observed that beams of particle such as electrons were diffracted from a crystal in a manner similar to the diffraction of x-rays, which are electromagnetic waves.

Experiment 2

By bombarding the material with photons the different energies, it was found that for each material a characteristic energy is required to remove the electrons from the surface of material. The minimum frequency of this electromagnetic radiation was found to be,

$$f = \frac{\phi}{h}$$
$$\phi \; is \; called \; work \; function \; of \; material$$

These two experiments proved that the sub-atomic particle possessed the properties of both wave as well as particle. Thus, wave-particle duality was developed.

Wave Function : Schrodinger's Equation

Since, there is no clear distinction between wave and particle in quantum mechanics, there should be some clear representation of what we call wave particle duality.

The displacement about its mean position is represented by wave-function ψ as
$\psi = Ae^{\frac{-j}{\hbar}[Et-px]}$ ---- (i)

Where, A is maximum displacement from mean position or amplitude

ħ is h/2π

E is energy of a particle

p is momentum of particle

x is displacement from origin

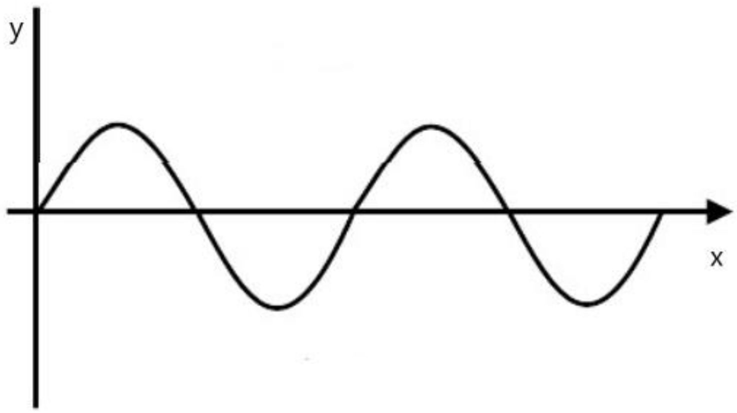


Fig : ordinary wave represented by above equation

Differentiating equation (i) w.r.t x

$\frac{\partial\psi}{\partial x} = Ae^{\frac{-j}{\hbar}[Et-px]} \cdot \frac{jp}{\hbar}$ ----- (ii)

Again differentiating equation (ii) w.r.t x

$\frac{\partial^2\psi}{\partial x^2} = Ae^{\frac{-j}{\hbar}[Et-px]} \left(\frac{jp}{\hbar}\right)^2$

$\frac{\partial^2\psi}{\partial x^2} = -\left(\frac{p}{\hbar}\right)^2 \psi$ ----- (iii)

For any system total energy E can be written as sum of kinetic energy and potential energy.

$E = \frac{p^2}{2m} + V$

$E - V = \frac{p^2}{2m}$

$p^2 = 2m(E - V)$ ----- (iv)

From equation (iii) and (iv)

$\frac{\partial^2\psi}{\partial x^2} = -\left(\frac{2m(E-V)}{\hbar^2}\right) \quad \psi$

$\frac{\partial^2\psi}{\partial x^2} + \left(\frac{2m(E-V)}{\hbar^2}\right) \quad \psi = 0$ ------ (v)

When V is time independent then E is also time independent then equation (v) is known as time independent S.W.E.

In 3D $\qquad \nabla^2 + \frac{2m[E-v]}{\hbar}\psi = 0$

Where, $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$

When V is time dependent then E is also time dependent called SWE.

$$\psi = Ae^{\frac{-j}{\hbar}[Et-px]}$$

Now, differentiating equation (i) wrt t we get,

$$\frac{\partial^2\psi}{\partial x^2} = -Ae^{\frac{-j}{\hbar}[Et-px]} \cdot \frac{jE}{\hbar}$$

$$\frac{\partial^2\psi}{\partial x^2} = -\frac{jE}{\hbar}\psi$$

$$E\psi = j\hbar\frac{\partial\psi}{\partial t}$$

Now, equation (v) can be written as,

$$\frac{\partial^2\psi}{\partial x^2} + \left(\frac{2mE}{\hbar^2}\right)\ \psi - \left(\frac{2mV}{\hbar^2}\right)\ \psi = 0$$

$$\frac{\partial^2\psi}{\partial x^2} - \left(\frac{2mV}{\hbar^2}\right)\ \psi = -\left(\frac{2m}{\hbar^2}\right)\ E\psi$$

$$\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} - V\psi = -j\hbar\frac{\partial\psi}{\partial t}$$

In 3D

$$\frac{\hbar^2}{2m}\nabla^2 - V\psi = -j\hbar\frac{\partial\Psi}{\partial t}$$

This is time dependent SWE.

Wave Function (ψ)

The wave can easily be understood and represented in classical mechanics. But the wave function in quantum mechanics does not have similar analogy. The wave function in quantum mechanics used to describe the observable quantities in quantum mechanics.

In quantum mechanics energy, momentum and position of the particle are called observables and the wave function ψ is used to describe these observables. These observables which characterize the particle cannot be measured at the same instant i.e only one can be measured at once.

The wave function is extensively used to predict the probability that a particle would be found in a specified volume of space. Wave function is a complex three dimensional quantity and is dependent upon the time. If ψ(x,y,z,t) represents the wave function then $|\psi|^2$ is called probability amplitude.

The probability of finding the particle in given incremental volume is equal to the probability amplitude (also called probability density function) and incremental volume i.e

| ψ(x,y,z)|^{2}dx dy dz

The probability of locating the particle between x and x+dx in single dimensional case, then the probability will be equal to | ψ(x)|^{2}dx.

Mathematically, for a particle located somewhere in the universe, we can write

$$\int_{-\infty}^{+\infty} \boldsymbol{\psi}\ (\mathbf{x},\mathbf{t})\mathbf{dx} = \mathbf{1}$$

- Schrodinger's equation does not tell us about the position of the particle rather it only gives us the probability of finding the particle in the vicinity of a point.
- A free particle is that which is not confined to any location. If potential energy V=0 in Schrodinger's equation for a particle in one-dimensional case is given by,

$$\boldsymbol{\Psi\Psi^*} = |\boldsymbol{A}|^{\mathbf{2}}$$

Electron in infinite potential well

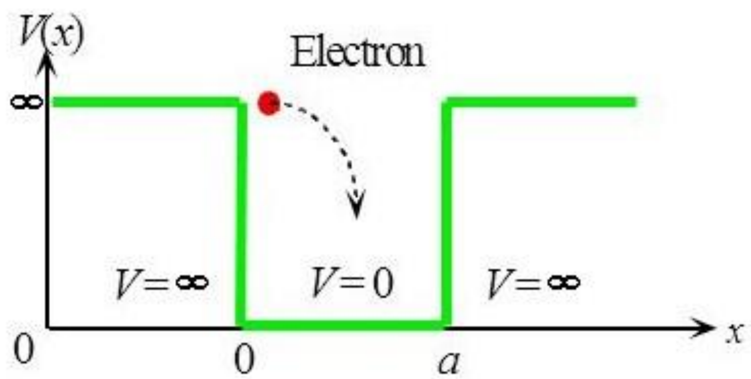


Fig : electron in infinite potential well

Let us consider an electron is present in the region 0<x<L where, a represents the width of the region in which electron is confined. Assume the potential energy of the particle is zero inside the confined region and infinite outside it. The electron thus confined cannot leave the region because to leave, it requires infinite potential energy.

Boundary condition for potential energy is given by

$$V = \begin{cases} 0 & 0 < x < L \\ \infty & 0 \geq x, L \leq x \end{cases}$$

Ψ is continuous function, so Ψ(0) = Ψ(L) = 0

Schrodinger's equation inside the box is given by,

$\frac{\partial^2\psi}{\partial x^2}+\left(\frac{2mE}{\hbar^2}\right)\quad\psi=0$

$\frac{\partial^2\psi}{\partial x^2}+k^2\psi=0$ ------- (i)

Where,

$k^2=\left(\frac{2mE}{\hbar^2}\right)$

General solution for equation (i)

$\psi(x)=Asinkx+Bcoskx$ …… (ii)

Where, A and B are some constants which we can find from boundary conditions.

We know from boundary;

$\psi(x)=0$ for $x=0$

here,

$0=A\cdot 0+B\cdot 1$

$B=0$

Again,

$\psi(x)=0$ for $x=L$

$0=ASinkL$

Since, $A\neq 0$ then $\sin kL=0=\sin n\pi$

When $n=1,2,3\ldots\ldots\infty$

$kL=n\pi$

$k=\frac{n\pi}{L}$

Now, from equation (ii)

$\psi(x)=A\sin\frac{n\pi}{L}x$

To find the value of A above equation can be normalized as;

$\int_0^L|\psi|^2dx=1$

$\int_0^L A^2\,sin^2\frac{n\pi}{L}x\,dx=1$

$\int_0^L\frac{A^2}{2}(1-\cos\frac{2n\pi}{L}x)dx=1$

$\frac{A^2}{2} x|_0^L = 1$

$A = \left(\frac{2}{L}\right)^{\frac{1}{2}}$

The normalized wave function of the electron in the box is;

$$\psi_n = \left(\frac{2}{L}\right)^{\frac{1}{2}} \sin\frac{n\pi}{L} x$$

This is called Eigen's function and value is used to measure the total energy of the particle inside the box.

$k^2 = \left(\frac{2mE}{\hbar^2}\right) = \frac{8\pi^2 mE}{h^2}$

$E = \frac{k^2 h^2}{8\pi^2 m}$

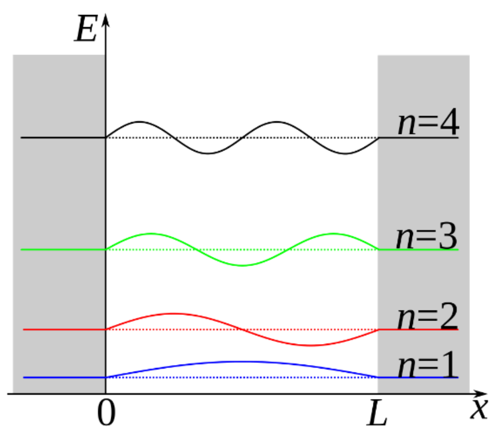


Fig: Energy level and wave function in the well

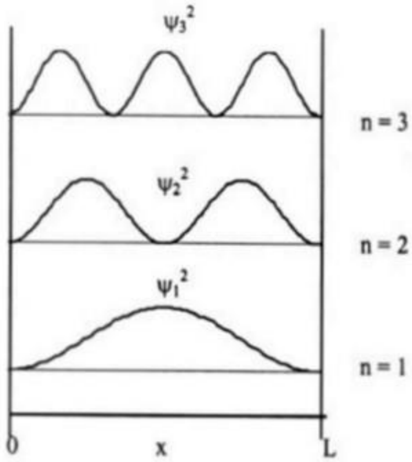


Fig. 3.2. The probability densities in One-dimensional particle-in-a-box

Finite Potential Barrier

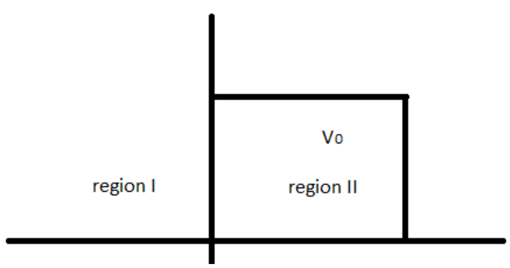


Fig : finite potential barrier

Consider a particle travelling in positive x-direction and encounters a barrier whose potential energy is V_0. Here, energy of the particle is less than the barrier potential energy also known as barrier height.

When applying Schrodinger's equation for the electron, here, we have to take into account the fact that there are two distinct region where the behavior of electron will be different.

In region 1st electron is free. So, it's potential energy is set equal to zero. So the Schrodinger's equation for region I will be,

$\frac{\partial^2 \psi_1}{\partial x^2} + \left(\frac{2mE}{\hbar^2}\right) \quad \psi_1 = 0$ -------- (i) for x<0

For region II, x>0

We can write Schrodinger's equation as

$\frac{\partial^2 \psi_2}{\partial x^2} + \left(\frac{2m(E-V_0)}{\hbar^2}\right) \quad \psi_2 = 0$ -----------(ii)

Above equations can be written as,

$\frac{\partial^2 \psi_1}{\partial x^2} + \beta^2 \psi_1 = 0$-------------- (iii)

$\frac{\partial^2 \psi_2}{\partial x^2} + \alpha^2 \psi_2 = 0$ ------------ (iv)

Where,

$$\beta^2 = \frac{2mE}{\hbar^2} \qquad \alpha^2 = \frac{2m(E-V_0)}{\hbar^2}$$

The solution of equation (iii) and (iv) can be written as,

$\psi_1 = Ae^{j\beta x} + Be^{-j\beta x}$ --------(v)

$\psi_2 = Ce^{j\alpha x} + De^{-j\alpha x}$ --------- (vi)

Where,

$Ae^{j\beta x}$ = wave travelling in +ve x-direction

$Be^{-j\beta x}$ = wave reflected from potential barrier

$Ce^{j\alpha x}$ = exponential increasing wave that penetrates the barrier in +ve x-direction

$De^{-j\alpha x}$ = exponential decreasing wave that penetrates the barrier

According to physics, interpretation ψ must be finite only and only if exponential increasing wave does not exists after the penetration of the barrier.

i.e. C=0

Now, the constant A, B and D in above equation (v) and (vi) can be found by using boundary condition.

At x=0, both the wave functions are continuous. So they must be equal to each other,

$\psi_1(0) = \psi_2(0)$

$Ae^{j\beta 0} + Be^{-j\beta 0} = De^{-j\alpha 0}$

$A + B = D$ -------- (vii)

The second boundary condition is the slope of both the wave function at x=0 are continuous

$\frac{\partial \psi_1}{\partial x} |_{x=o} = \frac{\partial \psi_2}{\partial x} |_{x=o}$

$jA\beta e^{j\beta x} - jB\beta e^{-j\beta x} = -jD\alpha e^{-j\alpha x}$

$jA\beta e^{j\beta 0} - jB\beta e^{-j\beta 0} = -jD\alpha e^{-j\alpha 0}$

$jA\beta - jB\beta = -jD\alpha$ -------- (viii)

From (vii) and (viii) we get,

$A = \frac{D}{2}\left[1 - \frac{\alpha}{\beta}\right]$

$B = \frac{D}{2}\left[1 + \frac{\alpha}{\beta}\right]$

According to the classical mechanics no particle can enter in region IInd but quantum mechanics proves that wave function has a finite value in region IInd wave function decreases exponentially in region IInd .

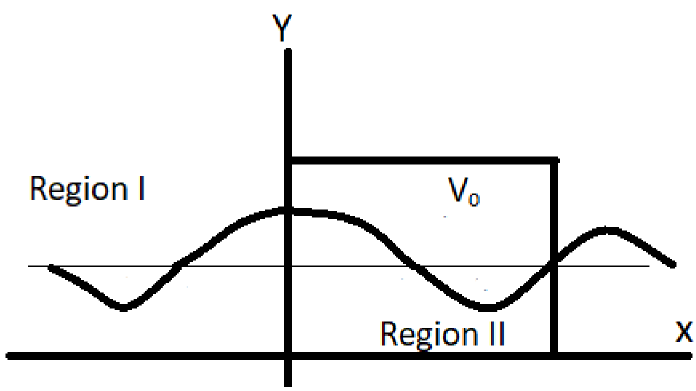


Fig : particle wave function in different region

Tunneling Effect

When beam of particle of energy 'e' is incident on the left of the potential barrier of height (V_0(V_0>E)). According to the classical mechanics the probability of any particle crossing the barrier is impossible. However, according to the quantum mechanics there is probability of the crossing the barrier. This behavior is called tunneling effect.

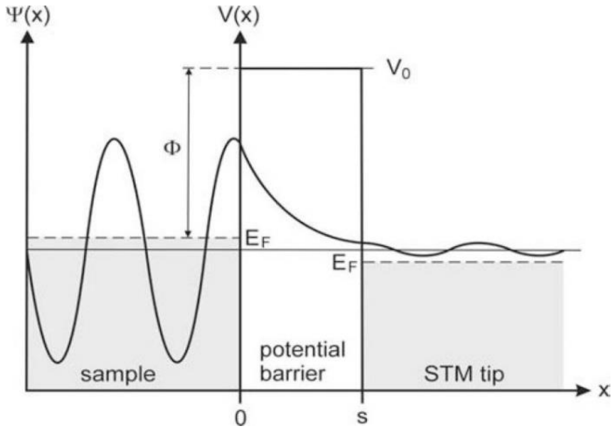


Fig : electron wave length in different region

In region 1, the Schrodinger's equation is

$\frac{\partial^2\psi_1}{\partial x^2} + \beta^2\psi_1 = 0$ -------- (i)

And the solution is,

$\psi_1 = Ae^{j\beta x} + Be^{-j\beta x}$ ------ (ii)

In region 2, the wave function is,

$\frac{\partial^2\psi_2}{\partial x^2} + \alpha^2\psi_2 = 0$ --------- (iii)

$\psi_2 = De^{-j\alpha x}$ ----------- (iv)

Similarly, in region 3, the Schrodinger's equation is,

$\frac{\partial^2\psi_3}{\partial x^2} + \beta^2\psi_3 = 0$ ----------- (v)

And the solution is,

$\psi_3 = Fe^{j\beta x} + Ge^{-j\beta x}$ ------------ (vi)

Where,

$\beta^2 = \frac{2mE}{\hbar^2}$

$\alpha^2 = \frac{2m(E-V_0)}{\hbar^2}$

$Ae^{j\beta x}$ = wave travelling in +ve x-direction

$Be^{-j\beta x}$ = wave reflected from potential barrier

$De^{j\alpha x}$ = exponential decrease wave that penetrates the barrier

$Fe^{j\beta x}$ = wave travelling in +ve x-direction onwards

$Ge^{-j\beta x}$ = wave reflected from region 3 which practically and theoretically does not exists = 0

The relative probability at the electron will turned from region 1 to region 3 through 2 is defined as transmission coefficient (T) which depends very strongly both on the relative potential barrier height (V_0-E) and the width of the potential barrier. The final expression for the transmission coefficient is expressed as,

$$T = \frac{|\psi_3(x)|^2}{|\psi_1(i)|^2} = \frac{F^2}{A^2}$$

$$T = \frac{1}{1+Dsinh^2(\alpha L)}$$

$$D = \frac{V_0^2}{4E(V_0-E)}$$

On solving we get,

$$T = \frac{16E(V_0-E)}{V_0^2} e^{-2\alpha L}$$

And the reflection coefficient R is given by,

$$R = \frac{B^2}{A^2} = 1 - T$$

Pauli Exclusion Principle

It states that “no two electrons in any one atom can have all four identical quantum number”.

Where,

n = principle quantum number

l = orbital angular momentum quantum number

m_l = magnetic quantum number

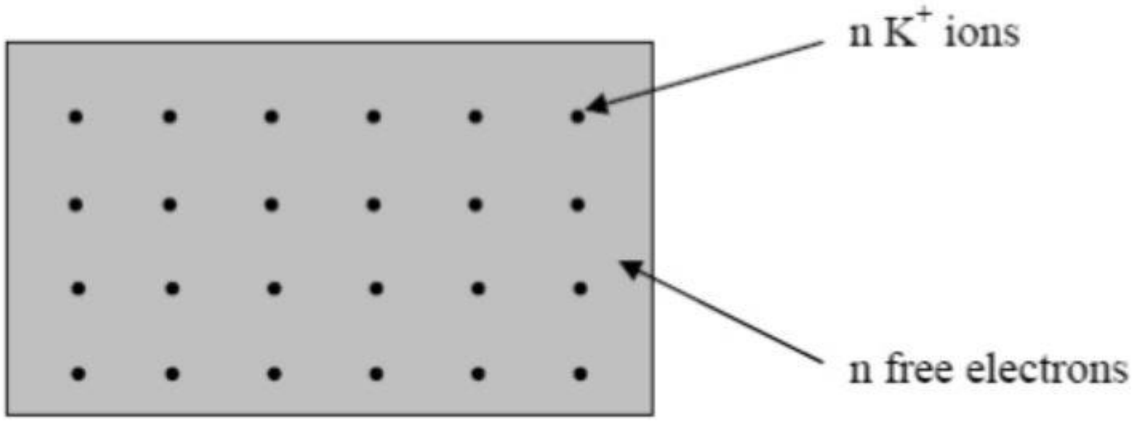
m_s = spin magnetic quantum number

and the wave function is given by, $\psi_{n,\mathrm{l},m_l,m_s,}$

According to the principle there must be at least one of the quantum number different than other in an electrons of an atom. If four quantum numbers of two electrons are identical, the one of the two electrons will be excluded from the electronic configuration of an atom. Therefore, this principle is called exclusion principle.

Free Electron Theory Of Metal

Drude in 1990 constructed theory of metal on the basis of kinetic theory of gases.



In gas a single type of particle is present but in metal at least two types of particles are present. i.e. electron and protons.

According to this theory the outermost electrons of metal known as valance electrons of atoms are loosely attached to the parent atom and these electron are free to move throughout the whole metal forming gas.

Momentum $(p) = mv$

Kinetic energy of electrons (KE)= $\frac{1}{2}mv^2 = \frac{p^2}{2m}$

We have,

$p = \hbar k$

$KE = \frac{(\hbar k)^2}{2m}$

$KE = ck^2$

Where, $c = \frac{\hbar^2}{2m}$

Fermi Energy

Fermi energy is defined as the highest filled energy level at a temperature of absolute zero. i.e.0^0K. All the energy levels up to the fermi level are filled at 0^0K and empty above it. The distribution of electron among various energy level in accordance with Pauli's exclusion principle. If N is the total numbers of electrons to be accommodated on the line then for even n, we have $N = 2n_f$. Thus, for $n = n_f$ the expression can be written as,

$E_F = \frac{\hbar^2}{2m}\left(\frac{\pi n_F}{2}\right)^2$ ------ (1)

Thus, the fermi energy depends upon the length of the box and the number of electrons in the box.

$E_F = \frac{\hbar^2}{2m}\left(\frac{\pi N}{2L}\right)^2 = \frac{\hbar^2}{2m}K_F^2$

Where, K_f = Fermi-wave vector

n_f = principle quantum number of the fermi level

The number of occupied electrons is sphere is given by;

$N = 2 \times \frac{Vol.\ of\ sphere\ of\ radius\ K_F}{Vol.\ of\ element\ of\ K-space}$

$N = 2 \times \frac{{}^4/_3\pi K_F^3}{K_x^3}$

$N = \frac{L^3K^3}{3\pi^2}$ $(K_x = \frac{2\pi}{L})$

$N = \frac{VK^3}{3\pi^2}$

And $n = \frac{N}{V} = \frac{K_f^3}{3\pi^3}$

Operators

Operators are used to predict the parameters like energy, momentum and position of an electron.

We have, SWE

$\psi = Ae^{\frac{-j}{\hbar}[Et-px]}$

Differentiating w.r.t t

$\frac{\partial\psi}{\partial t} = -Ae^{\frac{-j}{\hbar}[Et-px]} \cdot \frac{jE}{\hbar}$

$\frac{\partial\psi}{\partial t} = -\frac{jE}{\hbar}\psi$

$E\psi = j\hbar\frac{\partial\psi}{\partial t}$

This $E \rightarrow j\hbar\frac{\partial}{\partial t}$ is called energy operator.

Similarly, differentiating w.r.t x.

$$\frac{\partial\psi}{\partial x} = Ae^{\frac{-j}{\hbar}[Et-px]} \cdot \frac{jp}{\hbar}$$

Degenerate State

Different combination of principle quantum number representing the same energy is called degenerate state. The energy of electron in potential well is $E = \frac{n^2\hbar^2}{8mL^2}$. Here, principle quantum number 'n' is taken for single dimension and for three dimension, $n^2 = n_x^2 + n_y^2 + n_z^2$.

Then, the energy can be expressed as, $E_n = \frac{\hbar^2}{8mL^2} n_x^2 + n_y^2 + n_z^2$.

For same value of equation 'n' along x-axis can represent different energy because it also depends on 'n' along y and z-axis.

For example (1,2,1), (2,1,1) and (1,1,2) represent same electron energy. These all three electronic state are degenerate states.

Heisenberg's Uncertainty Principle

It is impossible to determine both position and momentum of an electron simultaneously with perfect accuracy. If an experiment is designate to measure one of them exactly, the other will become uncertain and vice-versa.

Let Δx be the uncertainty to introduce in measuring the position and Δp be the uncertainty to introduce in measuring momentum. If we try to measure both at once then according to uncertainty principle.

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Where, $\hbar = \frac{h}{2\pi}$, h Plank's constant.

Definition of Number Of States, Density Of States

The quantized energy state of electron is ;

$$E = \frac{n^2\hbar^2}{8mL^2}$$

$$E_n = \frac{\hbar^2}{8mL^2} n_x^2 + n_y^2 + n_z^2$$

Each set of quantum number exists a specific energy level called energy state. For n_x=n_y=n_z=1, there is one energy state. So unit cube has one energy state. Hence the number of energy state in any energy volume is proportional to that value.

The principle quantum number takes only positive value so it takes only radius ‘n’. The number of energy state within the sphere of radius ‘n’ is,

$$N = \frac{1}{8}\frac{4\pi n^3}{3}$$

Also,

$E = \frac{n^2\hbar^2}{8mL^2}$

$8mL^2E = n^2\hbar^2$

$n = \sqrt{\frac{8mL^2E}{\hbar^2}}$

Now,

$N = \frac{1}{8}\cdot\frac{4\pi}{3}\left[\frac{8\mathrm{mL}^2E}{\hbar^2}\right]^{3/2}$

$N = \frac{\pi}{6}\cdot K^{3/2}E^{3/2}$

Where,

$k = \frac{8mL^2}{\hbar^2}$

The number of state represent the number of energy level in quantum space. The density of states is number of energy state per unit energy in an energy interval of dE.

$\frac{dN}{dE} = \frac{3}{2}\frac{\pi}{6}K^{\frac{3}{2}}E^{\frac{1}{2}}$

$\frac{dN}{dE} = \frac{\pi}{4}K^{\frac{3}{2}}E^{\frac{1}{2}}$

Here, $\frac{dN}{dE} = Z(E) = Density\ of\ state$

When electron is placed in potential box with side ‘l’ then the volume of box is,

$V = l^3$

We have,

$Z(E) = \frac{dN}{dE} = \frac{\pi}{4}K^{\frac{3}{2}}E^{\frac{1}{2}}$

$Z(E) = \frac{dN}{dE} = \frac{\pi}{4}\left[\frac{8\mathrm{mL}^2}{\hbar^2}\right]^{3/2}E^{\frac{1}{2}}$

$Z(E) = \frac{dN}{dE} = \frac{\pi}{4}4\left[\frac{2m}{\hbar^2}\right]^{3/2}E^{\frac{1}{2}}V$

$Z(E) = \frac{dN}{dE} = 2\pi V \left[\frac{2m}{\hbar^2}\right]^{3/2} E^{\frac{1}{2}}$

$Z(E) = CE^{\frac{1}{2}}$

Where, $C = 2\pi V \left[\frac{2m}{\hbar^2}\right]^{3/2}$

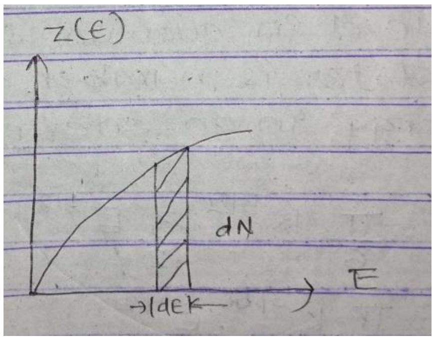


Fig: Density of state vs energy function

In the figure, the curve representing Z(E) is hyperbola where all energy levels less than E(n) are taken into account. The shaded area represents the number of energy state in an elementary energy interval d(E).

dN=Z(E)=dE

Contact Potential

Each metal has different work function which makes their fermi level different. When metal with different fermi level are in contact then electrons from metal with higher fermi level moves towards lower fermi level. This transfer of electron from higher fermi level to lower fermi level makes it positively charge and metals with low fermi level becomes negatively charge. Finally, a potential difference is developed at junction called contact potential. This process continues till the contact potential is large enough to prevent the further transfer of electron.

This is called system at equilibrium. The contact potential due to difference in work function can be calculated as;

$e\nabla v = \emptyset_2 - \emptyset_1$

$\nabla v = \frac{\emptyset_2 - \emptyset_1}{e}$

Thermal Emission And Work Function

The process of ejection of electrons from the surface of the metal at rather high temperature is called thermal emission. There is a threshold(minimum) value of energy which electron must possess if it is to leave the metal surface. The threshold value of energy is called work function of the metal. Work function is denoted by Greek alphabet ϕ.

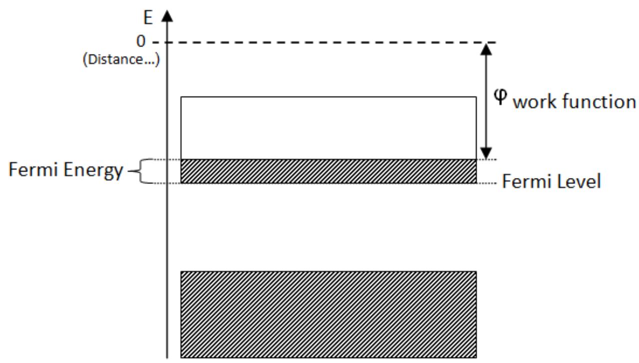


Fig: Fermi-level and work function in metal

In 1901 Richardson published the results of this experiment, the current from a heated wire seemed to depend exponentially on the temperature of the wire. A mathematical form similar to Arrhenius equation. Later he proposed that the emission law should have the mathematical form,

$$I = A_G T^2 e^{-\emptyset/KT}$$

I is emission current

T is temperature of the metal

ϕ is work function of the metal

K is Boltzman's constant

There was no consensus among interested theoreticians as to what is the exact expression for A_G but there is agreement that A_G must be written in the form,

$$A_G = \lambda_R A_0$$

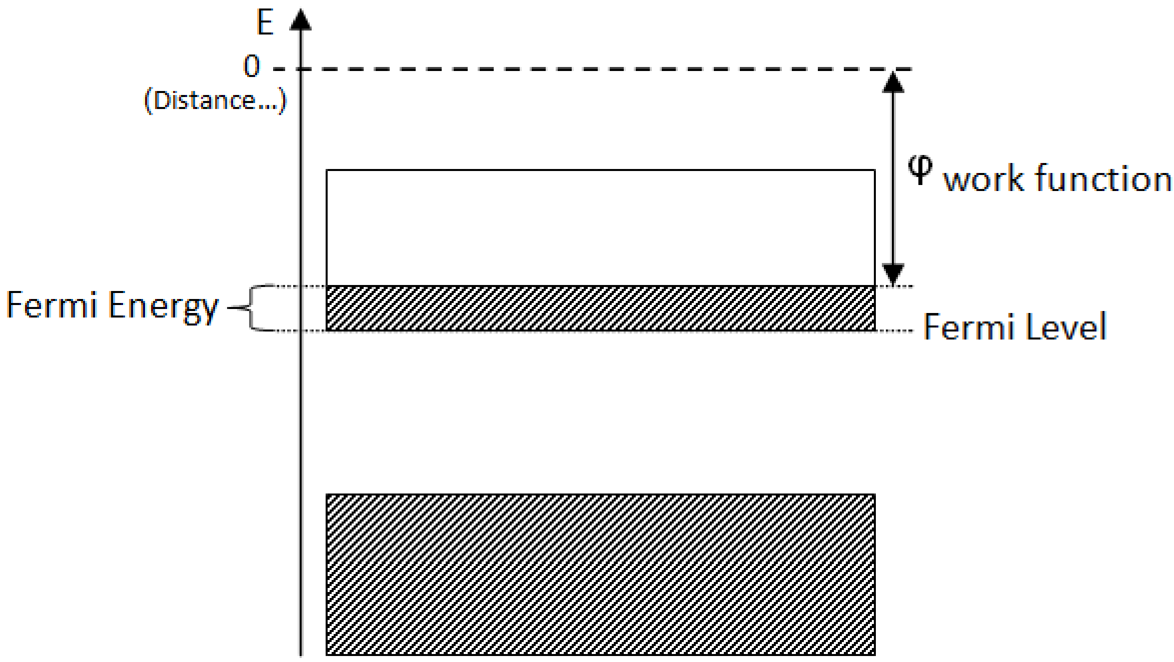
Where, λ_R is a material specific correction factor that is typically of order 0.5 and A_0 is a universal constant given by,

$$A_0 = \frac{4\pi m k^2 e}{h^3}$$

Where, m and e are the mass and charge of an electron and h is plank's constant.

Fermi-Dirac Distribution

Let us consider state of two particle with wave function ψ_1 and ψ_2 and their corresponding energy E_1 and E_2. This two particles goes on interaction or collision and they change their state.



Now, above particles have wave function ψ_3 and ψ_4 and their corresponding energy value E_3 and E_4 respectively after collision.

Now, after collision,

$f(E_1)f(E_2)[1-f(E_3)][1-f(E_4)] = f(E_3)f(E_4)[1-f(E_1)][1-f(E_2)]$------ (1)

$$Forward\ process = Backward\ process$$

$$\epsilon_1+\epsilon_2=\epsilon_3+\epsilon_4$$

Now, the fermi-Dirac distribution function is given by,

$F(E) = \frac{1}{1+e^{(E-E_F)/kT}}$

Where,

F(E) = Occupation index or probability of occupation

E = Energy of the electron

E_F = Fermi energy level

T = Absolute temperature

k = Boltzmann's constant

Case I : When T=0 and E_F>>E

$F(E) = \frac{1}{1+e^{-\infty}}$

$F(E) = 1$

Case II : When T=0 and E_F<<E

$F(E) = \frac{1}{1+e^{\infty}}$

$F(E) = 0$

Case III : When $T \neq 0$, E_f=E

$F(E) = \frac{1}{1+e^{0}} = \frac{1}{2}$

Case IV : When $T \neq 0$, E_f<E

$F(E) = \frac{1}{1+e^{E/kT}} = \frac{1}{e^{E/kT}}$

By using Fermi function or occupation index we can calculate the number of energy occupied by the electron.

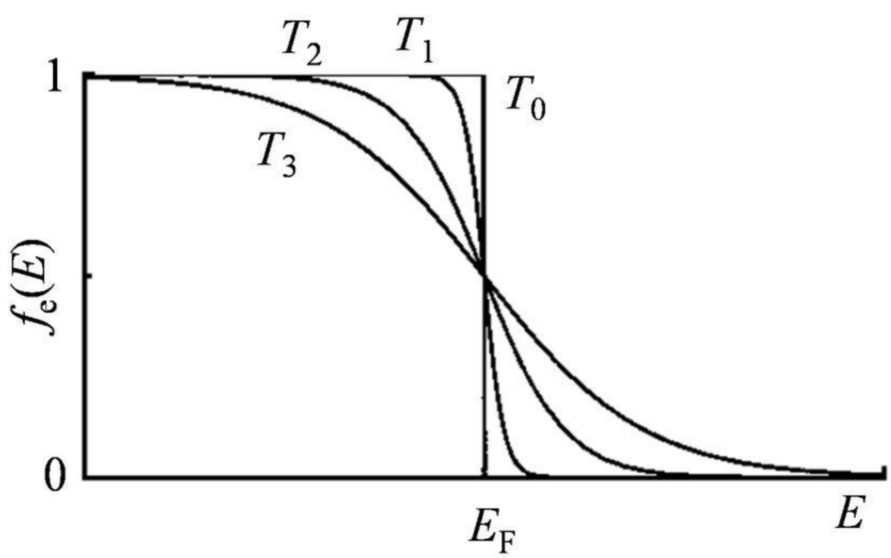


Fig : Fermi Dirac distribution